

Lab 3 Instructions

PUBH 523/623

Nicky Wakim

! Important

This lab is **ready!** (Nicky 8/10/26)

Directions

Please turn in your `.qmd` and `.html` file on Brightspace. Please use the same naming convention as practice assignments: `lastname_firstinit_lab_03.qmd` and `lastname_firstinit_lab_03.html`.

You can download the `.qmd` file for this lab [here](#).

The above link will take you to your **editing** file. **Please do not remove anything from this editing file!!** You will only add your code and work to this file.

Purpose

The purpose of this lab is to summarize your data and make a few more targeted data transformations that your summaries and future visualizations will need. In Lab 2, you filtered to your population, selected and renamed your variables, and built one or more composite scores. Now we'll actually look at what those variables tell us: quick numeric summaries, polished summary tables, and summaries grouped by a variable of interest. Along the way, you'll likely find that a variable isn't quite in the shape you need it in yet, that's the "further, needed data transformations" part of this lab. **This is meant to build your comfort translating a numeric result back into a plain-language sentence someone outside this class could understand.**

Lab activities

Restate your selected project

Recall the [following table from Lab 2](#) with each project's population, main variables, and composite score(s). You'll be summarizing the **main variables** and **composite score(s)** for your project throughout this lab.

#	Project Question	Population	Population variables	Main variables	Suggested additional variables	Composite score
1	For individuals under the federal poverty line, how do food support systems help individuals' food security?	People under 100% federal poverty line (FPL) and received free groceries or meals in the past 12 months	FPL_LT50, FPL50_100	FOOD_INSEC_Q23, PPETHM, Q37, Q26D_1, Q26D_2, Q26D_3, Q26D_4	NUM_CHILD_HH, Q25, Q25A, Q26, Q26A	Composite score of ease of access to free groceries or meals (combined score from Q26D_)

#	Project Question	Population	Population variables	Main variables	Suggested additional variables	Composite score
2	For individuals who identify as LGBTQ, how does trust and/or fair treatment in health-care effect self-reported health?	People who identify as LGBTQ and have a health-care provider. (You can change this population if you want. For example, you may want to focus on how trust and fair treatment may improve health for people with larger bodies. Check out Q70F for options!)	LGBTQ, Q108	Q108B_2, Q108B_1, Q27	Q108, Q70F_4, Q70F_5, PPETHM, Q280	Level of trust, courtesy, and respect around treatment (combined score from Q108B_1 and Q108B_2)

#	Project Question	Population	Population variables	Main variables	Suggested additional variables	Composite score
3	For individuals with school-aged children and below the federal poverty line, how do benefits received effect their mental health?	People who have at least one child (aged 5-18 years old) in the household and are under the 100% FPL	ANY_CHILDHH30_18, FPL_LT50, FPL50_100	Q30_18, Q30_2, Q30_3, Q30_4, Q30_5, Q30_6, Q53_A, Q53_B, Q53_C, Q53_D, Q53_E, Q53_G, Q53_H, Q53_I, Q53_J, Q11, Q31_4	NUM_CHILD_HH, Q1, Q37, PPETHM, Q27	Total mental health (combined scores from Q30_), Total benefits (combined scores from Q53_, Q11, and Q31_4)

! Task

Copy and paste the row from the above table that corresponds to your project into your `.qmd` file. This will help you (and me hehe) keep track of your project question, population, main variables, and composite score(s) as you work through this lab.

Import your saved data from Lab 2

! Task

From the `.rda` file you exported at the end of Lab 2, import your data into R using `import()` from the `rio` package. Save this as a new object called `wbns_05_composite`. Print out the number of rows and columns to verify the import.

Quick summaries of your variables

Before building any polished tables, get a feel for your variables using `skimr`.

! Task

Take your `wbns_05_composite` dataset and use `skim()` to generate a quick summary of your main variables and composite score(s). Save the result as a new object called `quick_summaries`, and print it out. Use this to check for any unexpected values, missing data, or other issues that may need to be addressed before moving forward.

Creating categorical versions of your composite score(s)

A continuous composite score is precise, but it's hard to summarize or visualize in a way that's easy to interpret at a glance (e.g., “the average score was 14.3” means less to a general audience than “most people reported easy access”). For this lab, create a **categorical version** of each composite score you built in Lab 2, so you have both forms available going forward.

There's no single correct set of categories — use your quick summaries above (the range, and where most of your data falls) to choose meaningful, roughly interpretable cut points. `case_when()` or `cut()` both work well for this.

Below are project-specific starting points. These are suggestions, not requirements — adjust the cut points and labels based on what your own data actually look like.

Project 1: `ease_free_food`

This composite ranges from 4 (very difficult across the board) to 20 (very easy across the board). A reasonable starting point is something like:

- 4-9: “Difficult access”
- 10-15: “Moderate access”
- 16-20: “Easy access”

Project 2: `trust_courtesy_respect`

This composite ranges from 2 (never true on both items) to 10 (always true on both). A reasonable starting point is something like:

- 2-5: “Low trust”
- 6-8: “Moderate trust”
- 9-10: “High trust”

Project 3: total_benefits

For `total_benefits`, you likely already have a natural, interpretable scale (a count of benefit types). A reasonable starting point is something like:

- 0-1: “0-1”
- 2-3: “2-3”
- 4-5: “4-5”
- 6-8: “6-8”

! Task

Use `mutate()` with `case_when()` or `cut()` to create a new categorical variable for each composite score, with meaningful category labels. Save the result as a new object called `wbns_06_cat`. Use `tabyl()` or `count()` to display how many respondents fall into each category, and confirm no one was left out (e.g., turned into an unexpected `NA`).

Building a summary table

Now put your quick summaries into a table that would make sense to someone outside this class (this is the kind of table you might include in your final information sheet). Use `tbl_summary()` from `gtsummary` to build one overall summary table of your composite score(s) and the other main variable.

Here are the specific variables to include by project:

- **Project 1:** `FOOD_INSEC` and the composite score of ease of access to free groceries or meals
- **Project 2:** `Q27` and the composite score of trust, courtesy, and respect around treatment
- **Project 3:** the composite score of total mental health and the composite score of total benefits

! Task

Use `tbl_summary()` to build a clean overall summary table of your composite score(s) and other main variable. Save this as a new object called `summary_table_overall`, and display it.

Customize at least one thing about the table’s appearance or labels (e.g., with `modify_header()` or by relabeling variables) so it’s easier for someone unfamiliar with your project to read.

Grouped summaries

Your project question is ultimately about a relationship (e.g., how ease of food access relates to food insecurity, or how trust relates to self-reported health). A single overall summary can't show that; you need to compare your composite score **across groups**.

For each project, we have a suggested grouping variable that you can use to stratify your summaries:

- **Project 1:** Use `FOOD_INSEC` to group your summaries of the composite score of ease of access to free groceries or meals.
- **Project 2:** Use `Q27` to group your summaries of the composite score of trust, courtesy, and respect around treatment.
- **Project 3:** Use your categorical total benefits to group your summaries of the composite score of total mental health.

! Task

Use `group_by()` and `get_summary_stats()` to calculate your composite score's common summary statistics separately for each level of your chosen grouping variable. Write 2-3 sentences interpreting what you see. Does the composite score noticeably differ across groups? Does this align with what you'd expect given your project question?

i Note

Optional/Additional: You can also pass your grouping variable into `tbl_summary(by = ...)` to get a nicely formatted side-by-side table instead of (or in addition to) the `group_by()/get_summary_stats()` output above.

Export tables

As always, save your work! Use `export()` in `rio` to save `wbns_06_cat` and your table `summary_table_overall` as a `.rds` file in your `data_project` folder. This is the dataset you'll build on for visualizations in Lab 4.

! Task

Save `wbns_06_cat` and `summary_table_overall` in a `.rda` file in your `data_project` folder. Name the file something that will help you identify it later, like `wbns_data_lab_03_work.rda` or `wbns_data_03_summ.rda`.